Cliques with Distributed Pendant Leaves: Symmetric Fractional Degree Bias

Abstract

Fix $r \geq 3$. Attach $n_i \geq 0$ pendant leaves to vertex i of K_r and put $h = n_1 + \cdots + n_r$. We prove that, whenever $1 \leq h \leq r$, every graphon W of edge density $p \geq 1 - 1/(r-1)$ satisfies

$$t(H, W) \geq p^h \prod_{j=1}^{r-1} \big(jp - (j-1)\big) = \Phi_H(p).$$

The leaf multiplicities may be distributed arbitrarily among the clique vertices. The proof averages over clique permutations, uses the arithmetic–geometric mean inequality to replace the distribution by the fractional degree weight $d^{h/r}$ at every clique vertex, and changes the underlying probability measure. A fractional Hölder inequality and a monotone rescaling of the accepted clique bound complete the argument. The case $(r, h) = (4, 2)$ with multiplicities $(1, 1, 0, 0)$ proves the formerly open NetworkX Atlas graph 134.

1 The graph family and chromatic target

Let (Ω, μ) be an arbitrary probability space and let $W \colon \Omega^2 \to [0, 1]$ be symmetric and measurable. Write

$$p = \int_{\Omega^2} W(x, y)\, \mathrm{d}\mu(x)\, \mathrm{d}\mu(y), \qquad d(x) = \int_{\Omega} W(x, y)\, \mathrm{d}\mu(y).$$

For integers $r \geq 3$ and $n_1, \ldots, n_r \geq 0$, let

$$H = H_r(n_1, \ldots, n_r)$$

be obtained from the clique on labelled vertices $1, \ldots, r$ by attaching n_i private pendant leaves to vertex i. Put

$$h = n_1 + \cdots + n_r.$$

The theorem below assumes $1 \leq h \leq r$; no restriction is placed on the individual n_i.

Since a leaf has $x - 1$ colour choices after its neighbour has been coloured,

$$\chi_H(x) = (x)_r (x-1)^h, \qquad (x)_r = x(x-1)\cdots(x-r+1). \tag{1}$$

The clique is a subgraph and the leaves do not increase chromatic number, so $\chi(H) = r$. Define the usual clique target

$$A_r(p) := \prod_{j=1}^{r-1} \big(jp - (j-1)\big).$$

Substituting $x = (1-p)^{-1}$ in (1) gives

$$\Phi_H(p) = p^h A_r(p). \tag{2}$$

Both sides are polynomials in p, so this identity includes the continued value 1 at $p = 1$. At the chromatic threshold

$$a_r := 1 - \frac{1}{r-1} = \frac{r-2}{r-1},$$

the final factor of A_r vanishes and hence $\Phi_H(a_r) = 0$.

We use exactly one previously accepted graphon theorem: for every graphon V of edge density $s \geq a_r$,

$$t(K_r, V) \geq A_r(s). \tag{3}$$

This is the Moon–Moser clique bound (Lemmas 2.2 and 2.3 of the paper). It applies to graphons on arbitrary probability spaces.

2 Fractional degree bias

We record the weighted-edge estimate needed after changing measure.

Lemma 2.1 (Fractionally degree-weighted edge)**.** *Let $0 < \alpha \leq 1$ and set*

$$N_\alpha = \int_{\Omega^2} W(x,y) d(x)^\alpha d(y)^\alpha \, \mathrm{d}\mu(x) \, \mathrm{d}\mu(y).$$

Then

$$N_\alpha \geq p^{1+2\alpha}.$$

Proof. On $\{d = 0\}$, define $W(x,y)/d(x)$ to be zero. Fubini's theorem implies that $W(x,y) = 0$ for almost every y there. Put

$$\theta = \frac{1}{1+2\alpha}, \qquad \eta = \frac{\alpha}{1+2\alpha}.$$

Thus $\theta + 2\eta = 1$ and, almost everywhere on the support of W,

$$W(x,y) = \big(W(x,y) d(x)^\alpha d(y)^\alpha\big)^\theta \left(\frac{W(x,y)}{d(x)}\right)^\eta \left(\frac{W(x,y)}{d(y)}\right)^\eta.$$

Weighted Hölder gives

$$p \leq N_\alpha^\theta \left(\int \frac{W(x,y)}{d(x)} \, \mathrm{d}\mu^2\right)^\eta \left(\int \frac{W(x,y)}{d(y)} \, \mathrm{d}\mu^2\right)^\eta.$$

Each quotient integral is the measure of a subset of $\{d > 0\}$ and is at most one. Hence $p \leq N_\alpha^\theta$, which is the assertion. □

The next scalar lemma is the reason that the upper bound $h \leq r$ is the natural boundary of this argument.

Lemma 2.2 (Clique rescaling)**.** *Let $r \geq 3$, let $1 \leq h \leq r$, and let $p \in (a_r, 1]$. Suppose $0 < z \leq p^h$ and put*

$$s_0 = p \left(\frac{p^h}{z}\right)^{2/r}.$$

If $s_0 \leq 1$, then

$$z A_r(s_0) \geq p^h A_r(p).$$

Proof. Every factor $js-(j-1)$ is positive for $s > a_r$. Put

$$\Psi_r(s) = \frac{A_r(s)}{s^{r/2}}.$$

For $s \in (a_r, 1]$, logarithmic differentiation gives

$$\begin{aligned}\frac{\Psi_r'(s)}{\Psi_r(s)} &= \sum_{j=1}^{r-1} \frac{j}{js-(j-1)} - \frac{r}{2s} \\ &= \frac{r-2}{2s} + \sum_{j=2}^{r-1} \frac{j-1}{s\big(js-(j-1)\big)} > 0.\end{aligned}$$

Thus Ψ_r is strictly increasing. The definition of s_0 is equivalent to

$$z = p^h \left(\frac{p}{s_0}\right)^{r/2}.$$

Since $s_0 \geq p$, it follows that

$$zA_r(s_0) = p^{h+r/2}\Psi_r(s_0) \geq p^{h+r/2}\Psi_r(p) = p^h A_r(p).$$

□

Also, A_r is nondecreasing on $[a_r, 1]$: every one of its factors is nonnegative and nondecreasing there.

3 The distributed-leaf theorem

Theorem 3.1 (Leaves distributed on a clique)**.** *Let $r \geq 3$ and let $n_1, \ldots, n_r \geq 0$ have total $1 \leq h \leq r$. For every graphon W of edge density*

$$p \in [a_r, 1],$$

one has

$$\boxed{t\big(H_r(n_1, \ldots, n_r), W\big) \geq p^h A_r(p) = \Phi_{H_r(n_1,\ldots,n_r)}(p).}$$

Proof. The claim is immediate at $p = a_r$, because the target is zero and every homomorphism density is nonnegative. At $p = 1$, one has $W = 1$ almost everywhere, and both sides equal one. Hence assume $p \in (a_r, 1)$.

For clique variables $x_1, \ldots, x_r$, write

$$F(\boldsymbol{x}) = \prod_{1 \leq i < j \leq r} W(x_i, x_j).$$

After integrating every leaf, the homomorphism density is

$$t(H, W) = \int_{\Omega^r} F(\boldsymbol{x}) \prod_{i=1}^{r} d(x_i)^{n_i} \, \mathrm{d}\mu^r.$$

Permuting the clique variables shows that the same integral is obtained after any permutation of the exponent vector $(n_1, \dots, n_r)$. Average over all $r!$ permutations. The arithmetic–geometric mean inequality, applied pointwise to the resulting $r!$ monomials, gives

$$t(H, W) \geq \int_{\Omega^r} F(\boldsymbol{x}) \prod_{i=1}^{r} d(x_i)^{h/r} \, \mathrm{d}\mu^r. \tag{4}$$

Indeed, each n_j appears in each coordinate exactly $(r-1)!$ times, so the geometric-mean exponent of every $d(x_i)$ is $(r-1)!h/r! = h/r$.

Set

$$\alpha = \frac{h}{r} \in (0, 1], \qquad M = \int_{\Omega} d(x)^{\alpha} \, \mathrm{d}\mu(x), \qquad z = M^r.$$

Since $p > 0$, one has $M > 0$. Define the probability measure

$$\mathrm{d}\nu(x) = \frac{d(x)^{\alpha}}{M} \, \mathrm{d}\mu(x).$$

The right side of (4) is

$$z \, t(K_r, W; \nu). \tag{5}$$

The edge density of W on (Ω, ν) is

$$s = \frac{N_{\alpha}}{M^2}.$$

Concavity of $u \mapsto u^{\alpha}$ gives

$$M \leq p^{\alpha}, \qquad 0 < z \leq p^h.$$

By Lemma 2.1,

$$s \geq \frac{p^{1+2h/r}}{M^2} = p \left(\frac{p^h}{z} \right)^{2/r} =: s_0.$$

Because $W \leq 1$ and ν is a probability measure, $s \leq 1$, so $s_0 \leq 1$. Also $s \geq s_0 \geq p > a_r$.

Apply the accepted clique bound (3) on the biased probability space, monotonicity of A_r, and Lemma 2.2:

$$\begin{aligned} t(H, W) &\geq z \, t(K_r, W; \nu) \\ &\geq z A_r(s) \geq z A_r(s_0) \geq p^h A_r(p). \end{aligned}$$

This is exactly (2). □

Every balanced complete k-partite graphon with $k \geq r-1$ gives the expected equality. At $k = r-1$ both sides vanish. For $k \geq r$, the degree is constant, equality holds in the symmetrization and fractional Hölder steps, the biased measure equals the original measure, and the accepted clique bound is sharp.

4 Atlas 134, audits, and exact limitations

Corollary 4.1 (Atlas 134)**.** *NetworkX Atlas graph* 134*, with graph6 code* `E~_O`*, is positive. It has order six, size eight, chromatic number four, and*

$$\chi_{F_{134}}(x) = x(x-1)^3(x-2)(x-3).$$

For every graphon of edge density $p \in [2/3, 1]$*,*

$$t(F_{134}, W) \geq p^3(2p-1)(3p-2) = \Phi_{F_{134}}(p).$$

Proof. Its NetworkX edge list is

$$01,\ 02,\ 03,\ 04,\ 12,\ 13,\ 23,\ 35.$$

Vertices $0, 1, 2, 3$ induce K_4, while 4 and 5 are leaves attached to the distinct clique vertices 0 and 3. Thus the graph is $H_4(1, 0, 0, 1)$, and Theorem 3.1 applies with $h = 2$. □

All integrations above concern bounded nonnegative measurable functions, so Tonelli–Fubini applies without an approximation or finite-step reduction. The quotient W/d is defined on $\{d = 0\}$ and the almost-everywhere vanishing needed there is stated explicitly. The biased measure is formed only after proving $M > 0$. The proof uses ordinary homomorphism densities, not injective copies, and assumes neither regularity nor a minimum-degree condition. The threshold and $p = 1$ endpoints are handled before any use of positive clique factors or logarithmic differentiation.

The theorem's exact new boundary is $h \le r$. This is what ensures that $d^{h/r}$ is a concave weight and that Lemma 2.1 applies with exponent at most one. The result does not cover more than r leaves distributed over several roots. It also requires the non-leaf core to be a complete clique and every attachment to be a pendant leaf; it is not a general leaf-attachment or mixed-chordal closure theorem.

The finite chromatic identities, scalar derivative, Hölder exponents, and Atlas identification were reconstructed in exact rational arithmetic. For formal verification, the reusable analytic pieces are the permutation symmetrization (4), the fractional edge lemma, and the scalar monotonicity of $A_r(s)/s^{r/2}$. The graph-specific module awaiting inspection is `lean/Taeyoung/Examples/Graph134.lean`.

5 What the Lean formalization actually proves

The machine-checked development is `lean/Taeyoung/Methods/CliqueDist/{Bias,Rows}.lean`, ending at `satisfiesLowerBound_134`. It proves the case $(r, h) = (4, 2)$ with multiplicities $(1, 1, 0, 0)$, and only that case. Three things are done differently.

Scope: $r = 4$, $h = 2$, and $\alpha = 1/2$. Theorem 3.1 is proved here for every $r \ge 3$ and every $1 \le h \le r$. The formalization fixes $r = 4$, $h = 2$. The reason is the exponent: $\alpha = h/r$ is $1/2$ there, so the biased density is $\sqrt{d}$ and every step stays inside 'Real.sqrt'; a general α would need 'Real.rpow' throughout and a fractional Hölder with r factors. No other member of the family is in the six-vertex catalogue.

The permutation average becomes a single exchange. Equation (4) averages the density over all $r!$ permutations of the exponent vector and applies AM–GM to the resulting $r!$ monomials. At $(4, 2)$ the formalization uses *one* exchange instead. Writing H_{ij} for K_4 with its two leaves at clique vertices i and j, the graphs H_{01} and H_{23} are isomorphic, so

$$2\,t(H, W) = \int F\big(d_0 d_1 + d_2 d_3\big) \ge \int F \cdot 2\sqrt{d_0 d_1 d_2 d_3} = 2 \int F \prod_i \sqrt{d_i},$$

where the middle step is the two-term AM–GM $a + b \ge 2\sqrt{ab}$ applied to the single perfect matching $\{01\}, \{23\}$. That produces exactly the geometric weight $\prod d_i^{h/r}$ of (4), from one graph isomorphism (`CliqueDist.isoExchange`, checked by `decide`) rather than an $r!$-fold average. The general case still needs the average; the matching shortcut is available whenever $h = r/2$ and the leaves sit on a perfect matching of the clique.

Lemma 2.2 becomes a factorization. This note proves the rescaling by showing $\Psi_r(s) = A_r(s)/s^{r/2}$ strictly increasing, through logarithmic differentiation and the identity

$$\frac{\Psi_r'}{\Psi_r} = \frac{r-2}{2s} + \sum_{j=2}^{r-1} \frac{j-1}{s(js-(j-1))} > 0.$$

At $r = 4$ the formalization uses no derivative. With $m = M^2$ and s the biased edge density, the two inputs are $m \le p$ and $ms \ge p^2$; then $m^2 s^2 \ge p^4$ and

$$p(2s-1)(3s-2) - s(2p-1)(3p-2) = (s-p)(6ps-2), \tag{6}$$

both of whose factors are nonnegative because $s \ge p \ge 2/3$. Multiplying out gives $m^2 A_4(s) \ge p^2 A_4(p)$ directly. Identity (6) is the $r = 4$ shadow of the monotonicity of Ψ_4, and it is a 'ring' step.

Everything else transcribes directly. Lemma 2.1 at $\alpha = 1/2$ is `GeometricMean.sq_le_integral_geoIntegran` already proved for `notes/two_root_triangle_leaf_cones.pdf`: it is precisely $N_{1/2} = \iint W\sqrt{d(x)d(y)} \ge p^2$, and the three-factor Hölder of this note's proof is the same one used there. The concavity bound $M \le p^\alpha$ becomes $M^2 \le p$, one application of the unweighted Cauchy–Schwarz `BookTail.sq_integral_mul_le` against the constant 1 — no concavity is needed. The change of measure (5) is `CliqueDist.integral_sqrtDegree_prod`, built on `Foundation/TiltTransfer.lean` exactly as `Methods/DegreeBias.lean` is at $\alpha = 1$. The clique input (3) is the accepted pure-chordal bound `cliqueDensity_ge_cliquePoly` at $r = 4$.